HUI Tools

User Manual

HUI-Display v1.2 • HUI-Test v1.9

A software toolkit for the Mackie HUI controller

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⚠ **WARNING:** DO NOT run HUI-Test at the same time as HUI-Display with Pro Tools connected. Running both simultaneously can cause erratic fader movement, display corruption, and loss of the Pro Tools–HUI connection. See Chapter 4 for details.

⚠ **WARNING:** HUI is a product of Mackie Designs. This software is an unofficial third-party utility and is not affiliated with, endorsed by, or supported by Mackie Designs or Avid Technology. Use at your own risk.

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Chapter 1 Introduction

1.1 What is HUI Tools?

HUI Tools is a small suite of software utilities for the Mackie HUI (Human User Interface) studio controller. It was created to solve a specific problem: the vacuum-fluorescent display (VFD) on a HUI that has failed, leaving the user unable to see plug-in parameters, dialog boxes, or other text that Pro Tools sends to that screen. HUI Tools intercepts and re-displays that information on a computer monitor, and also provides a comprehensive hardware test tool for diagnosing HUI problems.

The suite consists of two programs:

- **HUI-Display** — Intercepts the MIDI data that Pro Tools sends to the HUI's main display, shows it in a floating on-screen window, and passes all traffic through to the hardware unchanged so the rest of the unit continues to work normally.
- **HUI-Test** — Connects directly to the HUI hardware and lets you individually test every feature of the unit: LEDs, faders, V-pot rings, VU meters, the VFD display, relays, and the click and beep circuits.

1.2 The Mackie HUI

The Mackie HUI is a MIDI-controlled studio console surface manufactured by Mackie Designs in 1997. It communicates with Pro Tools (and compatible DAWs) via a MIDI connection, carrying fader positions, button presses, LED states, and display data back and forth. The main display is a 2-row, 40-character vacuum-fluorescent display (VFD) located in the meter bridge.

1.3 Package Contents

File	Description
hui_display.py	The HUI-Display program. Run this during Pro Tools sessions.
hui_test.py	The HUI-Test program. Run this for hardware diagnostics.
list_midi_ports.py	A small helper utility that lists all MIDI devices on your system.
config.hui	Shared settings file, created automatically on first launch.
SETUP_INSTRUCTIONS.txt	Quick-reference setup guide.

Chapter 2 Installation

You only need to complete this installation once.

> ⓘ **NOTE:** HUI-Test does not require loopMIDI. If you only intend to use HUI-Test, you can skip Step 2.

2.1 Step 1 — Install Python

- Go to https://www.python.org/downloads/ and download the latest Python.
- **CRITICAL:** On the first screen of the installer, tick "Add Python to PATH" before clicking Install Now.

2.2 Step 2 — Install loopMIDI (HUI-Display only)

- Go to https://www.tobias-erichsen.de/software/loopmidi.html and download loopMIDI.
- Launch loopMIDI. Create a port named HUI In and another named HUI Out.

> ✓ **TIP:** Enable "Autostart with Windows" in loopMIDI's Options so the ports are always available.

2.3 Step 3 — Install Python Libraries

```
python -m pip install --upgrade pip
pip install mido python-rtmidi
```

If the installation fails

If you see an error about "Unknown compiler" or "Visual Studio", download and install the Microsoft C++ Build Tools from https://visualstudio.microsoft.com/visual-cpp-build-tools/ (tick "Desktop development with C++"), restart your computer, then re-run the pip command above.

2.4 Finding Your HUI Port Names

Windows appends a number to every MIDI port name. To find the exact names, run:

```
python list_midi_ports.py
```

> ⓘ **NOTE:** Port numbers can change if you plug or unplug USB devices. Run list_midi_ports.py again to find the new numbers.

Chapter 3 HUI-Display

3.1 Overview

HUI-Display sits invisibly between Pro Tools and your HUI hardware, forwarding all MIDI traffic in both directions while watching for the specific SysEx messages that update the main 2×40 character VFD display. When those messages arrive, it decodes the text and shows it in a floating on-screen window.

The window renders the display in one of two modes:

- **Dot-Matrix mode (default)** — Renders each character as an accurate 5×7 pixel dot grid, replicating the look of the original VFD.
- **Text mode** — Renders the display as plain text using any monospace font installed on your system.

Every MIDI message passes through unchanged. Signal flow:

```
Pro Tools  →  [HUI In (loopMIDI)]  →  HUI-Display  →  HUI hardware
HUI hardware  →  HUI-Display  →  [HUI Out (loopMIDI)]  →  Pro Tools
```

3.2 First Launch and Configuration

- Turn on your HUI hardware and start loopMIDI.
- Double-click hui_display.py. Click Menu → Configure.

MIDI Ports tab

Port	What to select
Pro Tools → Script (input)	HUI In (the loopMIDI virtual port)
Script → HUI hardware (output)	Your physical HUI MIDI interface output port
HUI hardware → Script (input)	Your physical HUI MIDI interface input port
Script → Pro Tools (output)	HUI Out (the loopMIDI virtual port)

> **NOTE:** If your device does not appear in a drop-down, click Refresh port list.

Appearance tab

Controls the visual style of the display window. All interface elements use the Segoe UI font. Three colour pickers control the display appearance:

- **Lit pixel / text colour** — Defaults to a soft teal (#D1E9E4) that evokes the original VFD phosphor.
- **Background colour** — Defaults to near-black.
- **Labels and borders** — Status bar text and separator line colour.

Display mode

- **Dot-Matrix (default)** — Pixel-accurate 5×7 dot matrix rendering.
- **Text** — Plain monospace text. Faster and more readable at small sizes.

Dot-Matrix options

The Pixel size spinbox (1–16) controls how many screen pixels wide each VFD dot is.

Pixel size	Description
1	Pixel-perfect — one screen pixel per VFD dot. Smallest possible size.
2	Compact — no gaps between dots; clear and readable.
5 (default)	Standard — clear dot matrix at comfortable reading size on a 1080p display.
8–10	Large — suited to 4K monitors.

Text mode options

A Font drop-down lists common Windows monospace fonts. A Size spinbox sets the point size.

Font	Notes
Consolas (default)	Modern ClearType monospace font.
Courier New	Classic monospace, universally available.
Lucida Console	Wide, clear monospace.
Terminal / Fixedsys	Windows OEM bitmap fonts. Render sharpest at 9 pt.

Click Apply & Reconnect when done.

3.3 Configuring Pro Tools

- In Pro Tools, go to Setup → Peripherals → MIDI Controllers tab.

- Find your HUI entry and change the port settings:

> ⚠ **WARNING:** The terms "Send To" and "Receive From" are from Pro Tools' perspective. "Send To" is where Pro Tools sends data outward (to the HUI). "Receive From" is where it listens for data coming back.

Pro Tools setting	What to enter
Send To	HUI In ← Pro Tools outputs here; the script reads it
Receive From	HUI Out ← The script writes here; Pro Tools reads it

3.4 Daily Use

- Start loopMIDI, turn on your HUI hardware, double-click hui_display.py.
- Open Pro Tools. As soon as a session loads, the on-screen display populates with channel names and parameter values.

3.5 The About Dialog

Click About in the menu bar for version number, authorship, and licensing information.

3.6 Welcome Animation

When HUI-Display starts, a brief welcome animation plays on the on-screen VFD for approximately three seconds. A wipe effect fills both rows left to right with asterisks, then the name "HUI-Display" is typed out character by character, followed by the version number on the second row. The display then clears and switches to showing live Pro Tools data as normal.

> ℹ **NOTE:** The animation plays on the on-screen VFD window only. The connected HUI hardware is unaffected.

Chapter 4 HUI-Test

4.1 Critical Warning

⚠ **WARNING:** DO NOT run HUI-Test at the same time as HUI-Display with Pro Tools connected. HUI-Test connects directly to your HUI hardware over MIDI. If HUI-Display is simultaneously routing Pro Tools traffic through the same ports, both programs will compete for access, causing erratic fader movement, corrupted display output, Pro Tools losing its HUI connection, and loopMIDI auto-muting the virtual ports. Close HUI-Display before running HUI-Test, or disconnect the HUI in Pro Tools (Setup → Peripherals → MIDI Controllers).

4.2 Overview

HUI-Test is a diagnostic tool that communicates directly with the HUI hardware without Pro Tools. It is useful for verifying hardware operation, testing repairs, or learning how the HUI protocol works. Unlike HUI-Display, HUI-Test does NOT require loopMIDI.

4.3 Configuration

- Double-click hui_test.py and dismiss the startup warning.
- Click Menu → Configure → MIDI Ports and set the two port fields to your physical HUI hardware ports.
- Click Apply, then click the Connect button. HUI-Test zeros and blanks all hardware on connect, then displays a welcome message on the VFD.

ⓘ **NOTE:** The Appearance tab controls the colours and VFD Pixel Size for both HUI-Test and HUI-Display (via the shared config.hui file).

All interface elements in HUI-Test use the Segoe UI font. Data entry fields (VFD text boxes, scribble strip entries, timecode entry) use Courier New for accurate character-width alignment. The version number is shown in the About dialog (Menu → About) and in the welcome message displayed on the HUI VFD at connect time.

4.4 Connection Tab

Ping

Click Send Ping to send a ping message (note on, channel 0, note 0, velocity 0). If connected, the HUI will respond with a reply (velocity 127) within a few milliseconds.

Auto-ping

Enabled by default. Sends a ping every second to keep the HUI online. Without regular pings, the HUI goes offline after about two seconds and the faders stop responding.

MIDI Receive Log

Every MIDI message received from the hardware appears here in real time — button presses, fader touch events, fader positions, V-pot turns, and jog wheel movement.

Connection Timeout

If no ping reply arrives within five seconds of connecting, a pop-up error dialog appears with troubleshooting suggestions.

4.5 Live View Tab

Provides a real-time graphical display of all MIDI signals coming from the hardware. Press a button and watch it light up on screen; move a fader and see the bar rise; turn a V-pot and see the indicator move.

Graphical Canvas

The layout mirrors the physical HUI panel from left to right:

Section	Contents
Left control strip	Buttons for SELECT-ASSIGN, assignment controls, and channel/bank scroll arrows.
8 channel strips	Per-channel: fader touch indicator, V-pot ring, LED buttons (REC/RDY, INSERT, V-SEL, AUTO, SOLO, MUTE, SELECT), and fader percentage bar.
Transport section	All transport buttons, jog wheel indicator, and ping flash indicator.
Right panel	All remaining zones including KB Shortcuts, Window, DSP Param, Auto Enable, Auto Mode, Status/Group, Edit, Function Keys, Monitor In, Monitor Out, Timecode LEDs, and the three Numeric Keypad zones (Num Pad 1, 2, 3). Keypad buttons light up in Live View when pressed, even though the hardware has no LEDs for them.

Decoded Log

A scrollable log beneath the canvas shows every incoming event in plain English. A "Show raw MIDI" checkbox reveals the raw mido message string for each entry.

4.6 Display Tab

Main VFD (2×40 display)

Two text boxes correspond to the two 40-character rows. Type text and click Send to VFD. Clear VFD blanks both rows. Test pattern fills all positions with a repeating sequence.

Channel Scribble Strips

Nine 4-character entry boxes for the LED displays above the faders. Click Send All to update all nine simultaneously.

Timecode Display and Counter

The timecode entry box accepts HH:MM:SS:FF. The Timecode Counter animates the display automatically at a configurable frame rate — click Start Counter to begin, Stop Counter to halt.

Character Cycling

Cycles all printable characters through the display elements. Speed and Sequential offset controls adjust the rate and whether adjacent positions show different characters.

4.7 LEDs Tab

The HUI's LED system is organised into 29 zones, each with up to 8 ports.

Zone list and port buttons

Select a zone from the list to see its ports as toggleable buttons. Click any button to toggle the corresponding LED.

> ⓘ **NOTE:** Channel strip zones (0x00–0x07) show only ports 1–7 (Select, Mute, Solo, Auto, V-Sel, Insert, Rec/Rdy). Port 0 is the fader-touch sensor — it has no physical LED and does not appear in this panel. Zone 0x0D (Cursor / Mode / Scrub) shows only Mode, Scrub, and Shuttle, which are the only ports in that zone with LEDs.

All-On / All-Off

"Zone: All Ports ON/OFF" toggles all ports in the selected zone. "All LEDs ON/OFF (all zones)" does the same for every zone simultaneously.

LED Demo Modes

Mode	Description
Christmas Lights!	Each LED blinks independently at a random interval — chaotic and festive.

Chase	A single lit LED sweeps through every LED on the surface in sequence.
Strobe	All LEDs flash on and off in unison.
Heartbeat	All LEDs beat together in a rhythmic lub-dub pattern.

4.8 Meters and V-Pots Tab

VU Meters

Sliders set the level for each of the 16 meter columns (0 = off, 12 = all segments including clip). The Sweep toggle runs a phased sine-wave animation across all meters.

V-Pot Rings

Each of the 13 V-pots has a ring of 11 LEDs. Four display modes are available:

Mode	Description
Dot	A single LED at the specified position.
Pan	LEDs fill from the centre outward — left and right.
Level	LEDs fill from the left edge.
Spread	LEDs expand symmetrically from the centre.

The Centre LED checkbox independently controls the small indicator LED beneath each encoder knob. The V-Pot Sweep toggle animates all 13 rings simultaneously with phased offsets.

4.9 Faders Tab

Manual control

Eight vertical sliders drive the motorised faders directly. All to Top, All to Centre, and All to Bottom move all faders simultaneously.

⚠ **WARNING:** Moving faders via software will cause them to move physically. Ensure nothing is resting on or caught under any fader knob before sending fader commands.

Automated Motion Demo

Animates all eight faders in a repeating pattern. Five wave shapes: Sine (smooth), Square (fast jumps), Triangle (linear ramp), Sawtooth (ramp then snap), and Bounce (ball bounce). Speed, Amplitude, Centre, and Phase spread controls customise the motion.

4.10 Audio and Relays Tab

Relays (Zone 1D, Ports 0 and 1)

Two normally-open output relays for triggering external circuits (tally lights, On Air signs, etc.). Click the toggle button to open or close each relay.

Click (Zone 1D, Port 2)

The internal click speaker fires once per command and stops automatically. Single Click triggers it once; Click N times fires it in rapid succession.

Beeper (Zone 1D, Port 3)

The internal beeper sounds continuously until explicitly turned off. Always turn it off when done testing.

4.11 The VFD Display Window

Menu → View VFD Display opens a floating window rendering the current 2×40 VFD content as an accurate 5×7 dot-matrix pixel grid. It reflects whatever text HUI-Test has most recently sent — the welcome message, typed text, test patterns, or character cycle output.

> **NOTE:** This window shows what HUI-Test has sent, not necessarily what the physical VFD is displaying.

4.12 Goodies Tab

Vegas Mode

Click Start Vegas Mode to run a simultaneous multi-output animation that exercises every controllable element of the surface at once.

Output	Animation
Faders	All eight motorised faders follow phased sine waves at full 20 Hz update rate, producing smooth continuous rolling-wave motion.
VU Meters	All sixteen meter columns follow a counter-phased sine wave at full 20 Hz.
V-Pot Rings	All eight rings sweep in pan mode at 5 Hz with phased offsets.

LEDs	LEDs blink at slow random intervals (1–4 seconds per LED), one change per update cycle — a lazy, lounge-like Christmas Lights effect that minimises MIDI bandwidth consumption.
Main VFD	A scrolling banner moves continuously across both display rows.
Timecode	All eight timecode digits display random values each update, creating a slot-machine effect.

Click Stop Vegas Mode to halt all animations. On stop, all outputs are zeroed and blanked.

⚠ **WARNING:** It may take up to 5 seconds after clicking Stop Vegas Mode for all outputs to zero. This is a consequence of MIDI bandwidth limits and is normal.

ⓘ **NOTE:** Vegas Mode does not control the relays, beeper, or click circuit.

4.13 Test Wizard

The Test Wizard provides a guided, sequential test of all detectable HUI hardware features. It is the fastest way to do a comprehensive health-check of the surface after a repair or when diagnosing a fault.

ⓘ **NOTE:** The HUI must be connected before starting the wizard. Click Connect in the Connection tab first.

Overview

Click Run Test Wizard to begin. A progress bar and step counter show your position in the sequence. Click Stop at any time — results logged so far remain visible and exportable. Click Step Selection… before starting to choose which sections and button groups to include.

Step types

Type	Behaviour
Auto	Waits for a physical hardware action and detects the resulting MIDI. Passes automatically when the expected input is received. Skips on timeout.
Manual	Sends a stimulus to the hardware and asks you to inspect the result. Pass and Fail buttons appear. A countdown auto-skips if you do not respond.

Ready → Manual	Used for tests requiring a safety check before proceeding (e.g. fader motor sweep). An "I'm Ready" button is shown first. Click it to confirm it is safe to proceed; the wizard then fires the test and shows Pass/Fail buttons.
Button Group	All buttons in a group are lit simultaneously. Press each button in turn — its LED turns off (or the beeper sounds for no-LED buttons) when detected. Continue advances when all are pressed or timeout expires.
Section	A divider between test groups. Auto-advances after a brief pause. No result is recorded.

Test sections

Connection

Sends a ping and waits for the HUI to reply. Confirms basic MIDI communication.

Faders — Move & Touch

One move step and one touch step per fader (16 steps total, faders 1–8). The move step detects any position change greater than ~3% of travel. The touch step detects the capacitive touch sensor. The strip LEDs above the fader being tested flash to indicate which fader is active.

V-Pots — Rotate encoders

One step per encoder — all 13 V-pots: channels 1–8, ASSIGN 1–4, and the SCROLL encoder. Each step detects rotation in either direction.

Jog Wheel

Detects at least three clockwise jog wheel clicks.

Buttons

All surface buttons are tested in 21 grouped steps. Within each group, all button LEDs are lit simultaneously. Press each button — its LED turns off when detected.

Group	Buttons covered
Channel Strip — Ch 1	Select, Mute, Solo, Auto, V-Sel, Insert, Rec/Rdy for channel 1
Channel Strip — Ch 2	Same seven buttons for channel 2
Channel Strip — Ch 3–4	All 14 channel-strip buttons for channels 3 and 4
Channel Strip — Ch 5–6	All 14 channel-strip buttons for channels 5 and 6

Channel Strip — Ch 7–8	All 14 channel-strip buttons for channels 7 and 8
Keyboard Shortcuts	CTL, SHF, Edit Mode, Undo, Alt, Opt, Edit Tool, Save
Window Buttons	Mix, Edit, Transport, Mem-Loc, Status, Alt
Channel / Bank Scroll	← CH, ← BK, CH →, BK →
Assignment 1	Output, Input, Pan, Send E, D, C, B, A
Assignment 2	Assign, Default, Suspend, Shift, Mute, Bypass, Rec All
Cursor / Mode / Scrub	Down *, Left *, Mode, Right *, Up *, Scrub, Shuttle (* = beep feedback; no LED)
Transport	Talkback, Rewind, Fast Fwd, Stop, Play, Record
Transport — Loop / RTZ	RTZ, End, On Line, Loop, Quick Punch
Transport — Punch / Audition	Audition, Pre, In, Out, Post
Numeric Keypad	0–9, ., +, −, ×, ÷, =, CLR, Enter (all: beep feedback; no LEDs)
Auto Enable	Plug-in, Pan, Fader, Snd Mute, Send, Mute
Auto Mode	Trim, Latch, Read, Off, Write, Touch
Status / Group	Phase, Monitor, Auto, Suspend, Create, Group
Edit	Paste, Cut, Capture, Delete, Copy, Separate
Function Keys	F1 through F8/ESC
Parameter Edit	Ins/Para, Assign, Select 1–4, Bypass, Compare

NOTE: Buttons marked with * have no physical LED. A short beep from the HUI's internal beeper confirms detection of each such button press.

Output Tests

Manual tests that send stimuli to hardware outputs and ask for your visual or auditory confirmation.

Step	Stimulus sent	What to check
VFD Display	Fills both rows with "HUI-Test**" repeating — all 80 character positions, no blank spaces.	Both rows fully lit end-to-end with no blank or garbled positions.
Scribble Strips	Writes TST1–TST9 on each of the nine scribble strip displays.	All nine strips readable and correct.

LEDs	Turns on every LED on the surface simultaneously.	All LEDs lit; note any that remain dark.
V-Pot Rings	Sets all 13 rings to full: all 11 ring LEDs plus the centre indicator on simultaneously.	Every LED in every ring lit, including the centre dot beneath each knob.
VU Meters	Drives all 16 meter columns to maximum (all 12 segments including clip LED).	All segments lit on all 16 columns.
Fader Motors	I'm Ready prompt, then sweeps all 8 faders bottom → top → bottom. The VFD shows "Faders moving, please wait…" during the sweep, then "Sweep done — confirm Pass or Fail" when complete.	All faders moved smoothly and reached both ends of travel.
Timecode Counter	All 8 digits cycle through 0–9 independently at different speeds, like a slot machine.	All eight digits visibly changing; no stuck or dark segments.
Click	Fires the click circuit 20 times rapidly.	Rapid clicking audible from the HUI speaker.
Beeper	Sounds the beeper 5 times — 500 ms on, 500 ms off per pulse.	Five distinct beep pulses audible.
Relay 1	Opens and closes Relay 1 five times at 0.25-second intervals.	Five distinct clicks audible; external equipment activates if connected.
Relay 2	Same as Relay 1, for Relay 2.	As above for Relay 2.

⚠ **WARNING:** During the Fader Motors step an "I'm Ready" confirmation is required before the faders move. Ensure nothing is resting on or caught under any fader knob before clicking I'm Ready.

Step Selection dialog

Click Step Selection… to open a two-column dialog showing all test sections with checkboxes. Within the Buttons section, each of the 21 button groups has its own indented sub-checkbox.

- Tick or untick any section to include or exclude it entirely.
- Tick or untick individual button groups beneath the Buttons section to fine-tune which groups are tested.
- Disabling the Buttons section header does not reset individual group checkboxes — you can toggle the section on and off without losing per-group selections.

- Select All and Select None operate on all checkboxes at all levels simultaneously.

> ⓘ **NOTE:** The Connection step is always included and cannot be disabled.

VFD display during the wizard

While the wizard is running, the HUI main VFD shows the current step on row 1 and a brief instruction on row 2 (for example "Waiting for hardware…" for auto steps, "Press each lit button" for button groups). At the end of each step, the word PASS, FAIL, or SKIP appears at the right-hand end of row 2 for approximately one second before the display advances to the next step.

> ⓘ **NOTE:** The VFD test step is excluded from this behaviour — it writes its own test content to the display.

Results log and export

Symbol	Meaning
✔ PASS	The expected input was detected (auto / btn_group step) or you clicked Pass.
✘ FAIL	You clicked Fail, or a btn_group step ended with buttons not pressed.
— SKIP	Step timed out or you did not respond to a manual step.

A summary line at the end shows totals. The hardware is then reset to the zeroed state. Click Export Report at any time to open a Save As dialog and choose where to save the report and what to name it. The default filename includes a timestamp so successive reports do not overwrite each other.

Chapter 5 The config.hui File

config.hui is a shared settings file used by both programs, stored in JSON format in the same folder as the program files. A typical config.hui:

```
{
  "pt_sends_to":         "HUI In 3",
  "hui_out_port":        "(your MIDI interface output port)",
  "hui_in_port":         "(your MIDI interface input port)",
  "pt_recv_from":        "HUI Out 5",
  "fg_color":            "#d1e9e4",
  "bg_color":            "#060a08",
  "dim_color":           "#003d2c",
  "font_size":           15,
  "pixel_size":          5,
  "display_mode":        "dot_matrix",
  "text_font":           "Consolas",
  "skip_test_warning":   false
}
```

Key	Meaning
pt_sends_to	loopMIDI port that Pro Tools sends to (HUI-Display only)
hui_out_port	Physical HUI hardware MIDI output port
hui_in_port	Physical HUI hardware MIDI input port
pt_recv_from	loopMIDI port that Pro Tools receives from (HUI-Display only)
fg_color	Lit pixel / text colour (hex code)
bg_color	Window background colour (hex code)
dim_color	Label and border colour (hex code)
pixel_size	Dot-matrix pixel size 1–16 (screen pixels per VFD dot)
display_mode	"dot_matrix" (default) or "text"
text_font	Font name for text mode (e.g. Consolas)
font_size	Font size in points for text mode
skip_test_warning	If true, suppresses the HUI-Test startup warning

NOTE: To reset all settings to defaults, delete config.hui. Both programs recreate it with defaults on next launch.

Chapter 6 Troubleshooting

"MIDI ERROR" in the status bar

- Open Configure → MIDI Ports, click Refresh port list and re-select your ports.
- Check that loopMIDI is running (HUI-Display only).
- Run list_midi_ports.py to see current port names and numbers.

The display shows blank text

- Make sure Pro Tools is running with a session open.
- Check that the "display SysEx received" counter in the status bar is incrementing.
- Verify Pro Tools peripheral settings per Chapter 3.3 and reconnect if necessary.

loopMIDI has muted a virtual port

loopMIDI mutes ports when it detects what it thinks is a feedback loop. This is usually a false alarm during the initial Pro Tools handshake. To unmute: open loopMIDI, click the red/orange port, and click Reset. Disable the automatic muting threshold in loopMIDI's Options to prevent recurrence.

Port numbers have changed

Unplug/replug events change Windows MIDI port numbers. Run list_midi_ports.py and update the configuration.

The HUI goes offline after a few seconds

The HUI requires a ping every two seconds. Make sure Auto-ping is enabled in the Connection tab (HUI-Test) or that Pro Tools is actively communicating (HUI-Display).

The program crashes on startup

Run from Command Prompt to see the error message:

```
python hui_display.py
python hui_test.py
```

python-rtmidi installation fails

See Section 2.3. Install Microsoft C++ Build Tools and retry.

Chapter 7 Limitations

7.1 MIDI-Only Interface

HUI Tools communicates exclusively over MIDI. This covers all normal operational functions but not hardware-level diagnostics. Faults below the MIDI layer — failed LED drivers, VFD filament failure, power supply issues — cannot be observed over MIDI and will not be detectable by HUI-Test.

7.2 VFD Display Limitations

Internal ROM messages

Some VFD content is generated by the HUI's own firmware without any MIDI command — including the startup splash screen and internal diagnostic messages. HUI-Display cannot capture or reproduce these.

Adjustment dialogs

Certain configuration dialogs (talkback routing, muting options, dim attenuation) are produced entirely by the HUI's internal firmware and are never sent over MIDI. HUI-Display cannot show them.

> **NOTE:** These firmware dialogs function normally on the physical VFD. HUI-Display does not interfere with them.

7.3 Hardware-Level Diagnostics

HUI-Test can only confirm that a MIDI command was sent; it cannot confirm whether the hardware physically responded. A command may succeed at the MIDI level while the physical hardware does nothing. Always observe the physical unit directly alongside the software display when diagnosing faults.

Chapter 8 Licensing

HUI Tools Software Licence

HUI Tools (HUI-Display and HUI-Test) is © 2026 Richard Philip and is released as **free, open-source software** under the **GNU General Public License v3.0** (GPL-3.0).

You are free to run, study, redistribute, and improve the software, provided that any distributed versions remain under the GPL-3.0. The complete licence text is available at:

https://www.gnu.org/licenses/gpl-3.0.html

> i **NOTE:** The GPL applies to the software only. This manual is separately licensed under CC BY-NC-SA 4.0.

Third-Party Acknowledgements

Library	Details
mido	MIDI Objects for Python. MIT Licence. https://mido.readthedocs.io/
python-rtmidi	Python binding for RtMidi. MIT Licence. https://github.com/SpotlightKid/python-rtmidi
tkinter	Standard Python GUI library (PSF Licence). Part of the Python standard library.

HUI Protocol

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